Yashi Huang

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SUMMARY

Operations Management PhD candidate specializing in mathematical optimization, simulation, and probabilistic forecasting. Integrates predictive modeling with optimization to support scheduling and capacity planning, with hands-on experience deploying machine-learning systems in clinical operations.

EDUCATION

University of Rochester, Ph.D. Candidate in Operations Management 09/2021 – 12/2026 (Expected)

- GPA: 3.9/4.0

Columbia University, M.S. in Operations Research 09/2018 – 02/2020

- GPA: 3.9/4.0

Wuhan University, B.A. in Financial Engineering 09/2014 – 06/2018

- GPA: 3.8/4.0

TECHNICAL SKILLS

- **Optimization & OR:** Mixed-integer programming, Gurobi, scheduling, simulation, queueing, resource allocation, scenario analysis.
- **Machine Learning & Statistics:** Random Forest, XGBoost, neural networks, probabilistic forecasting, time-series modeling, survival analysis, PCA, statistical inference, hypothesis testing, model calibration and validation.
- **Programming & Platforms:** Python (pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow), SQL, R, MATLAB, VBA; PySpark, Databricks, Snowflake; automated testing, CI/CD, version control.

PROFESSIONAL EXPERIENCE

Data Scientist Intern, Dana-Farber Cancer Institute – Boston, MA May 2025 – Aug 2025

- Designed scalable Python, PySpark, and Databricks/Delta Lake data pipelines processing 1M+ patient records; engineered appointment, queue-state, and provider-level features from multiple operational sources.
- Advanced a wait-time prediction prototype into a production machine-learning system for clinical operations; evaluated probabilistic forecasts and prediction-interval reliability.
- Implemented MLflow-supported model evaluation, automated retraining, drift monitoring, and post-deployment reporting to support reliability and maintainability.
- Collaborated with 15+ clinicians, product managers, and engineers to align model outputs with workflow needs; communicated findings through dashboards and stakeholder presentations.

Quantitative Researcher, Veta Investment Partners – Topeka, KS Sep 2020 – Apr 2021

- Developed and backtested quantitative equity-trading signals using Python, SQL, statistical models, and volatility/momentum features to evaluate investment strategies.
- Automated SQL/Python portfolio analytics and performance reporting, saving approximately four hours per day; communicated risks and returns through dashboards.

SELECTED RESEARCH PROJECTS

Probabilistic Intraday Demand Forecasting & Capacity Planning Oct 2024 – Present

- Studied a general probabilistic set-to-curve forecasting problem, learning set functions that map arbitrary collections of inputs to distributions of time-varying outcomes.
- Introduced an aggregation—disaggregation framework combining machine learning, time-series models, and simulation to generate point and quantile forecasts across organizational levels.
- Conducted a comprehensive empirical comparison of machine-learning, statistical, and simulation-based forecasting methods using real-world hospital data, assessing nominal and quantile predictive performance.
- Investigated forecasting performance through a simulation-based study with oracle benchmarks and replicated training datasets, disentangling irreducible uncertainty from modeling error and training-sample variability.
- Built a Python and Excel/VBA decision-support tool to assess how changes in staff schedules affect predicted occupancy, peak demand, and overtime.

Predictive Scheduling in Appointment-Based Service Systems Oct 2024 – Present

- Framed predictive scheduling as estimating a feasible realized service schedule under stochastic arrivals, service durations, and no-shows, extending beyond independent appointment-level predictions.
- Designed a simulation-optimization framework incorporating Dynamic Time Warping (DTW) to measure discrepancies between variable-length service timelines and identify representative schedules.
- Proposed a mixed-integer linear programming (MILP) formulation to minimize weighted timing deviations across simulated realizations, subject to service-order, arrival-precedence, and non-overlap constraints.
- Evaluated the framework against representative-scenario and heuristic benchmarks across different operational uncertainty settings, assessing prediction accuracy and schedule feasibility.

Dynamic Waiting-Time Prediction & Uncertainty Quantification Mar 2023 – Present

- Formulated real-time waiting-time prediction as a conditional survival problem, updating probabilistic estimates as queue and provider information evolve.
- Customized the interval-quantile selection procedure combined with split conformal prediction to construct calibrated prediction intervals that balance coverage–width tradeoff.
- Established XGBoost's performance advantage on evaluated metrics using statistical comparisons of Random Survival Forests, XGBoost, and neural networks.
- Developed a nested simulation framework to quantify irreducible prediction error and estimate achievable forecasting performance under system uncertainty.

SELECTED WORKING PAPERS

Huang, Y., Senderovich, A., and Shaposhnik, Y. *Dynamic Interval Prediction of Patient Waiting Times in a Cancer Care Facility*. Working Paper, 2026.

Huang, Y., Senderovich, A., and Shaposhnik, Y. *Forecasting Infusion Chair Occupancy as a Probabilistic, Curve, Set Function Prediction*. Working Paper, 2026.

INVITED TALKS

Oct 2025 "Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility," INFORMS Annual Meeting, Atlanta, GA.

Jul 2024 "Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility," INFORMS MSOM Conference, Minneapolis, MN.

Aug 2023 "Physician Rostering Problem with Downstream Capacity Constraints," INFORMS Healthcare Conference, Toronto, CA.